

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 7

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Due 2026-10-18 23:59

Topic

Conservation Laws and the Newtonian Limit

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Conservation Laws and the Newtonian Limit. State all assumptions and conventions.

$$J_{ai} = \partial R_a / \partial \theta_i$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\alpha_1 = -1/2$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$\alpha_2 = -1/4$$

Show enough intermediate work that another student can reproduce the calculation.